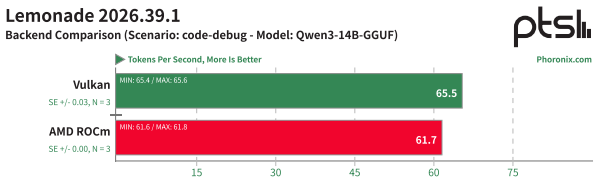
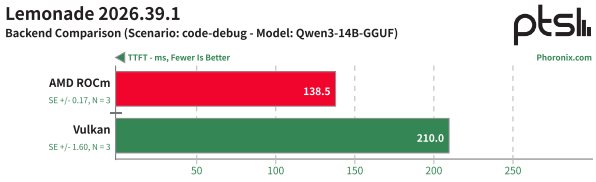
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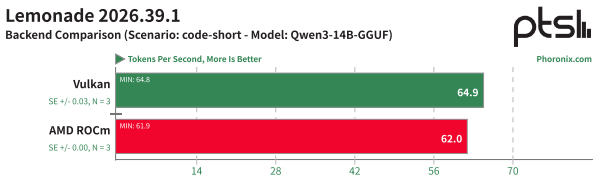
# AMD ROCm vs. Vulkan Performance For Lemonade Local AI Server With Llama.cpp

Written by Michael Larabel in Display Drivers on 5 October 2026 at 09:50 AM EDT. **Page 3 of 3**. 15 Comments.

And then firing up the System76 Thelio Major with Ryzen Threadripper and AMD Radeon AI PRO R9700 graphics, to see if it was the same story as the Strix Halo APU or different.




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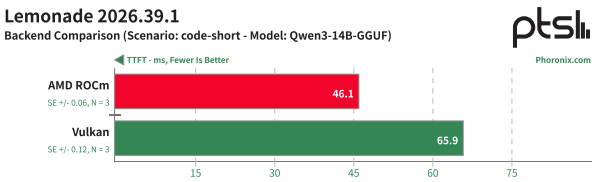
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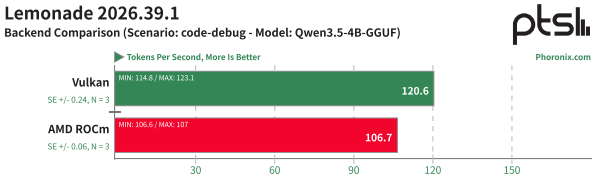
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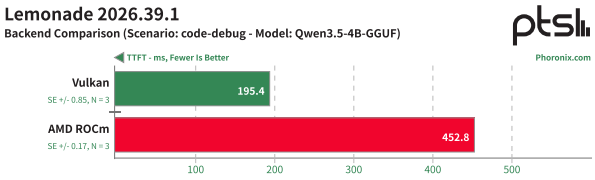
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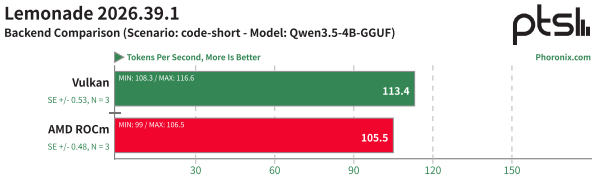
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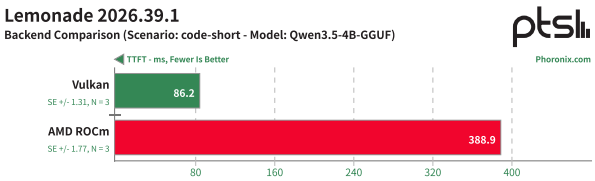
Framework Laptop 12 Offers Improved Performance & Efficiency With Intel Core 3

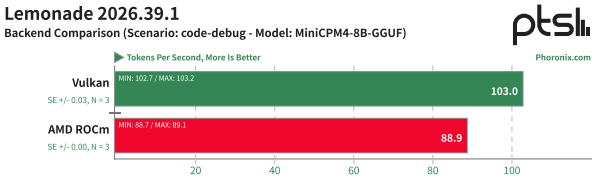


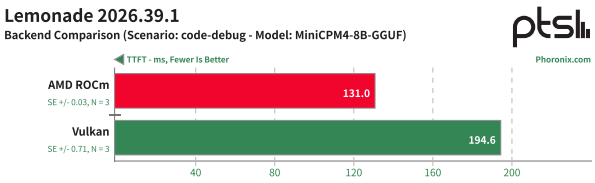




Even with the completely different hardware and different Ubuntu release and kernel, it was the same story as seen on the Framework Desktop: the Vulkan back-end was delivering more tokens per second while the ROCm back-end enjoyed lower latency.

Wildcat Lake

ROCm 10 OpenCL vs. Mesa 26.3-dev Rusticl Performance On AMD Strix Halo

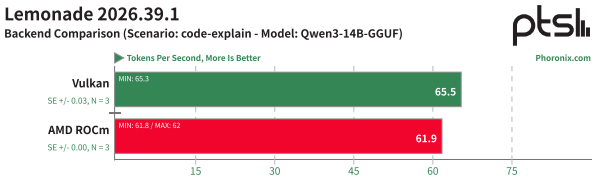
AMD PerfOpt Boosting Prior-Gen Ryzen APU Performance For AI

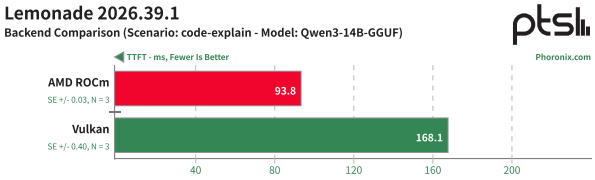
Ubuntu 26.10 Delivers Nice I/O Performance Wins Over Ubuntu 26.04 On AMD EPYC

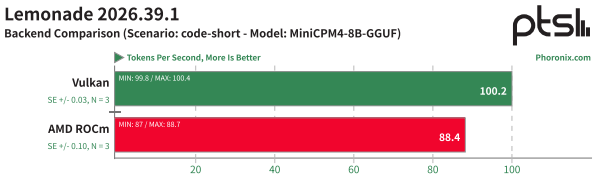
Star Labs StarFighter Ultra: A Modern High-End Linux Laptop With Coreboot

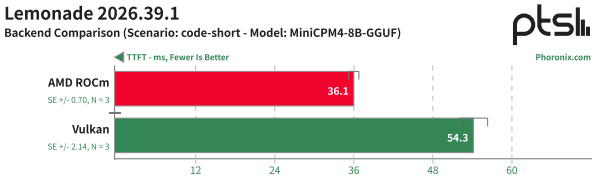
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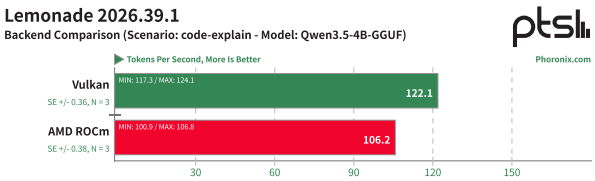
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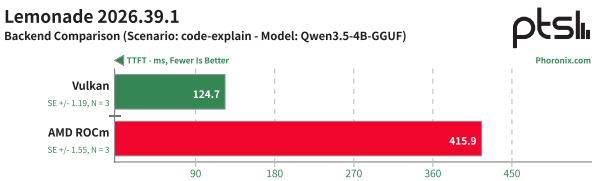

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-14B-GGUF)
Tokens Per Second, More Is Better
Phoronix.com
Vulkan
SE +/- 0.03, N = 3
MIN: 65.3
65.5
AMD ROCm
SE +/- 0.00, N = 3
MIN: 61.8 / MAX: 62
61.9
15
30
45
60
75

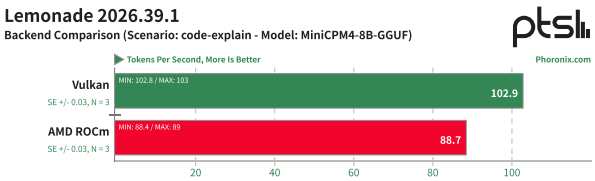

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-14B-GGUF)
TTFT - ms, Fewer Is Better
Phoronix.com
AMD ROCm
SE +/- 0.03, N = 3
93.8
Vulkan
SE +/- 0.40, N = 3
168.1
40
80
120
160
200

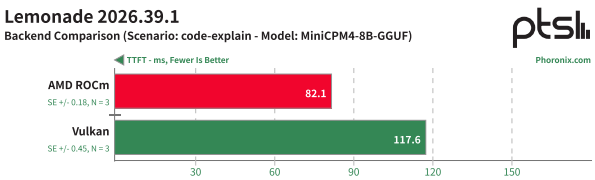

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: MiniCPM4-8B-GGUF)
Tokens Per Second, More Is Better
Phoronix.com
Vulkan
SE +/- 0.03, N = 3
MIN: 99.8 / MAX: 100.4
100.2
AMD ROCm
SE +/- 0.10, N = 3
MIN: 87 / MAX: 88.7
88.4
20
40
60
80
100

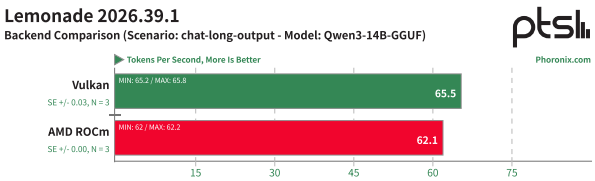

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: MiniCPM4-8B-GGUF)
TTFT - ms, Fewer Is Better
Phoronix.com
AMD ROCm
SE +/- 0.70, N = 3
36.1
Vulkan
SE +/- 2.14, N = 3
54.3
12
24
36
48
60

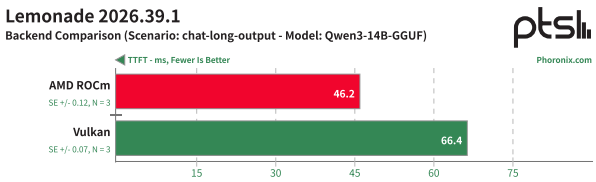

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3.5-4B-GGUF)
Tokens Per Second, More Is Better
Phoronix.com
Vulkan
SE +/- 0.36, N = 3
MIN: 117.3 / MAX: 124.1
122.1
AMD ROCm
SE +/- 0.38, N = 3
MIN: 100.9 / MAX: 106.8
106.2
30
60
90
120
150


Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3.5-4B-GGUF)
TTFT - ms, Fewer Is Better
Phoronix.com
Vulkan
SE +/- 1.19, N = 3
124.7
AMD ROCm
SE +/- 1.55, N = 3
415.9
90
180
270
360
450


Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: MiniCPM4-8B-GGUF)
Tokens Per Second, More Is Better
Phoronix.com
Vulkan
SE +/- 0.03, N = 3
MIN: 102.8 / MAX: 103
102.9
AMD ROCm
SE +/- 0.03, N = 3
MIN: 88.4 / MAX: 89
88.7
20
40
60
80
100

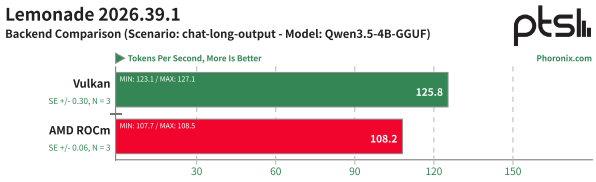

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: MiniCPM4-8B-GGUF)
TTFT - ms, Fewer Is Better
Phoronix.com
AMD ROCm
SE +/- 0.18, N = 3
82.1
Vulkan
SE +/- 0.45, N = 3
117.6
30
60
90
120
150

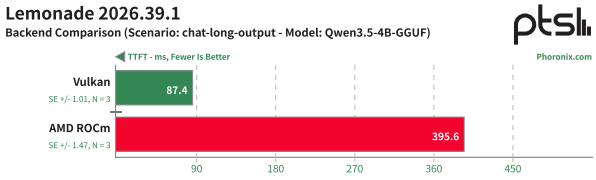

Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: Qwen3-14B-GGUF)
Tokens Per Second, More Is Better
Phoronix.com
Vulkan
SE +/- 0.03, N = 3
MIN: 65.2 / MAX: 65.8
65.5
AMD ROCm
SE +/- 0.00, N = 3
MIN: 62 / MAX: 62.2
62.1
15
30
45
60
75

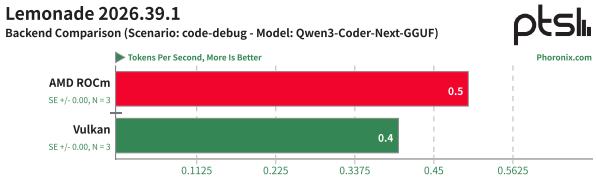


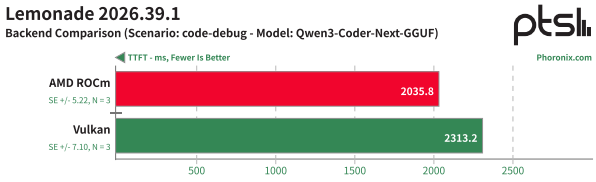

The time to first token with ROCm tended to be quite significant while the Vulkan throughput advantage was typically more incremental . In a few cases the Vulkan back-end with the Radeon AI PRO R9700 did enjoy faster times to first token compared to the ROCm back-end.

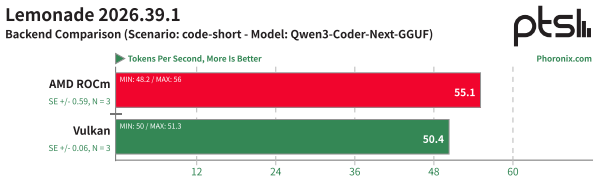
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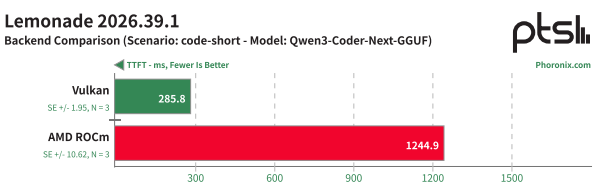


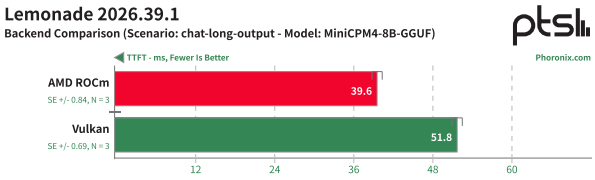



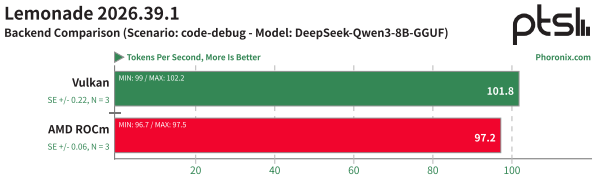



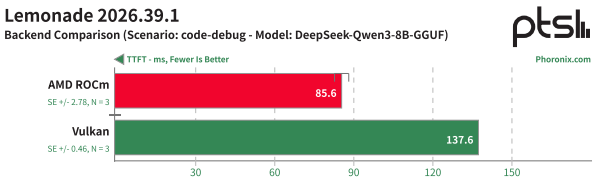



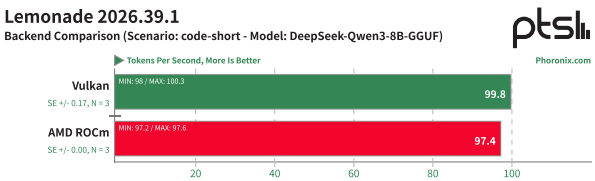

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
Vulkan
SE +/- 1.95, N = 3
285.8
AMD ROCm
SE +/- 10.62, N = 3
1244.9
300
600
900
1200
1500

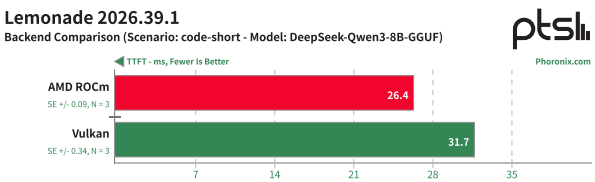

Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: MiniCPM4-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.06, N = 3
MIN: 103.4 / MAX: 103.9
103.6
AMD ROCm
SE +/- 0.15, N = 3
MIN: 87.1 / MAX: 89.4
89.1
20
40
60
80
100

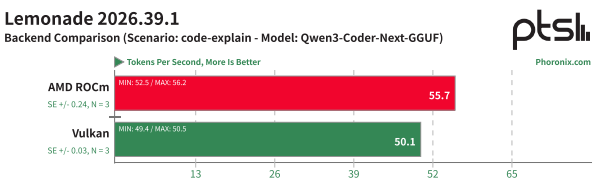

Lemonade 2026.39.1
Backend Comparison (Scenario: chat-long-output - Model: MiniCPM4-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.84, N = 3
39.6
Vulkan
SE +/- 0.69, N = 3
51.8
12
24
36
48
60

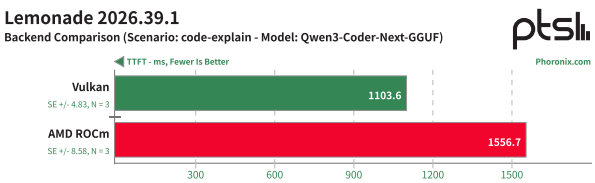

Lemonade 2026.39.1
Backend Comparison (Scenario: code-debug - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.22, N = 3
MIN: 99 / MAX: 102.2
101.8
AMD ROCm
SE +/- 0.06, N = 3
MIN: 96.7 / MAX: 97.5
97.2
20
40
60
80
100

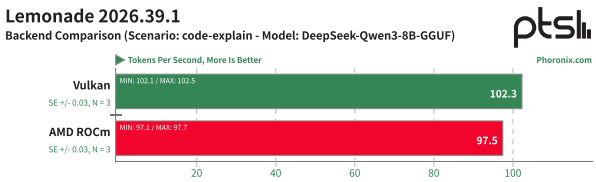

Lemonade 2026.39.1
Backend Comparison (Scenario: code-debug - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 2.78, N = 3
85.6
Vulkan
SE +/- 0.46, N = 3
137.6
30
60
90
120
150

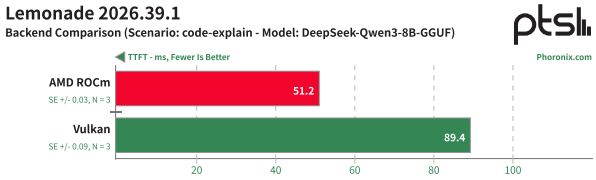

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
Vulkan
SE +/- 0.17, N = 3
MIN: 98 / MAX: 100.3
99.8
AMD ROCm
SE +/- 0.00, N = 3
MIN: 97.2 / MAX: 97.6
97.4
20
40
60
80
100

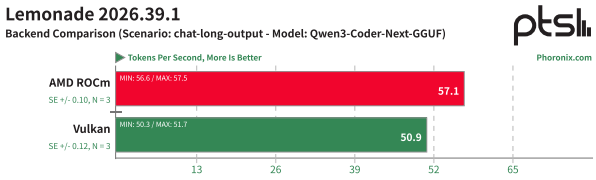

Lemonade 2026.39.1
Backend Comparison (Scenario: code-short - Model: DeepSeek-Qwen3-8B-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
AMD ROCm
SE +/- 0.09, N = 3
26.4
Vulkan
SE +/- 0.34, N = 3
31.7
7
14
21
28
35

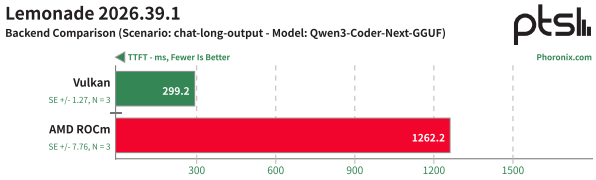

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
Tokens Per Second, More Is Better
AMD ROCm
SE +/- 0.24, N = 3
MIN: 52.5 / MAX: 56.2
55.7
Vulkan
SE +/- 0.03, N = 3
MIN: 49.4 / MAX: 50.5
50.1
13
26
39
52
65

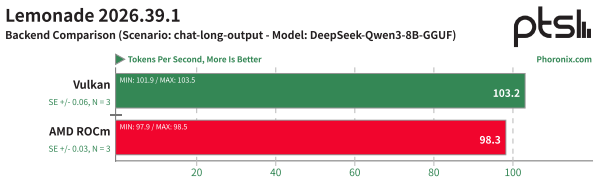

Lemonade 2026.39.1
Backend Comparison (Scenario: code-explain - Model: Qwen3-Coder-Next-GGUF)
Phoronix.com
TTFT - ms, Fewer Is Better
Vulkan
SE +/- 4.83, N = 3
1103.6
AMD ROCm
SE +/- 8.58, N = 3
1556.7
300
600
900
1200
1500

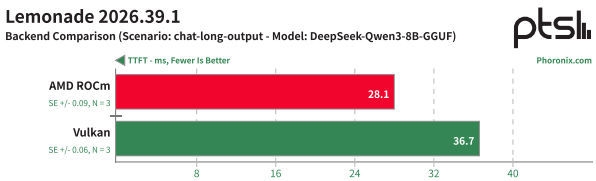



So for those wondering, the Vulkan back-end of Llama.cpp is still working quite well overall and working across GPU vendors. The AMD ROCm back-end though as shown on both the Radeon AI PRO R9700 and Strix Halo Radeon 8060S Graphics was providing lower latency performance.

## 15 Comments

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## About The Author

Michael Larabel is the principal author of Phoronix.com and founded the site in 2004 with a focus on enriching the Linux hardware experience. Michael has written more than 20,000 articles covering the state of Linux hardware support, Linux performance, graphics drivers, and other topics. Michael is also the lead developer of the Phoronix Test Suite, Phoromatic, and OpenBenchmarking.org automated benchmarking software. He can be followed via Twitter, LinkedIn, or contacted via MichaelLarabel.com.



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